

DEEP LEARNING

Labwork 3 - Logistic Regression

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1 Algorithm Implementation

In this lab, we implemented logistic regression from scratch to predict whether a person would get a loan based on their experience and salary. The goal is to understand the effect of gradient descent optimization and the role of the learning rate in convergence.

1.1 Model Definition

The logistic regression model predicts the probability of a positive outcome using the sigmoid function:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = w_1 x_1 + w_2 x_2 + w_0$$

1.2 Loss Function

The binary cross-entropy loss for N data points is:

$$J(w_0, w_1, w_2) = \frac{1}{N} \sum_{i=1}^N L_i = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

1.3 Gradients

The partial derivatives of the loss function with respect to the weights are:

$$\frac{\partial J}{\partial w_0} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)$$

$$\frac{\partial J}{\partial w_1} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i) x_i^{(1)}$$

$$\frac{\partial J}{\partial w_2} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i) x_i^{(2)}$$

1.4 Gradient Descent

We update the weights iteratively:

$$w_j \leftarrow w_j - r \frac{\partial J}{\partial w_j}, \quad j = 0, 1, 2$$

where r is the learning rate.

2 Effect of Learning Rate

The learning rate r significantly affects convergence:

- **Small learning rate** ($r = 0.001$): Very slow convergence, many iterations needed.
- **Moderate learning rate** ($r = 0.05$): Fast and stable convergence.
- **Large learning rate** ($r = 0.1$): May overshoot minimum, loss oscillates, may diverge.

Choosing an appropriate learning rate is crucial for gradient descent efficiency and stability.

3 Result Visualization

The graph shows the logistic regression decision boundary for predicting loan approval based on Experience and Salary. Red points indicate profiles approved for a loan, blue points indicate profiles refused, and the green line represents the decision boundary learned by gradient descent.



Figure 1: Logistic Regression graph